

Chuhao Zhou

(86)138-8181-3524 | zhouchuhao99@gmail.com | github.com/ChuhaoZhou99

EDUCATION

School of Mechanical and Aerospace Engineering, Nanyang Technological University

PhD Candidate

Aug 2024 - Present

- **Awards:** NTU Research Scholarship (2024).

School of Computer Science, Harbin Institute of Technology

Master of Engineering

Sep 2021 - Jan 2024

- **GPA:** 3.72/4.00 (Top 3% in 60 students).
- **Awards:** National Scholarship (2023, Top 1% in 176 students), HIT 1st Academic Scholarship*3 (2020-2021/2021-2022/2022-2023, top 5% in the department)).

School of Computer Science, Harbin Institute of Technology

Bachelor of Engineering

Aug 2017 - Jul 2021

- **GPA:** 3.91/4.00 (Top 3% in 118 students).
- **Awards:** HIT 2nd Academic Scholarship*3 (2017-2018/2018-2019/2019-2020, top 10% in the department), Outstanding Graduate Award of HIT (2021, top 5% in the department).

PUBLICATIONS

- [1] **Chuhao Zhou**, Jianfei Yang. HoloLLM: Multisensory Foundation Model for Language-Grounded Human Sensing and Reasoning. *Advances in Neural Information Processing Systems (NeurIPS'25)*. 2025.
- [2] **Chuhao Zhou**, Yuzhe Zhou, Tingting Ren, Huafeng Li, Jinxing Li, and Guangming Lu. Hierarchical disturbance and Group Inference for video-based visible-infrared person re-identification. *Information Fusion*. 2024.
- [3] **Chuhao Zhou**, Jinxing Li, Huafeng Li, Guangming Lu, Yong Xu, and Min Zhang. Video-based Visible-Infrared Person Re-Identification via Style Disturbance Defense and Dual Interaction. *ACM International Conference on Multimedia (MM'23, Oral)*. 2023.
- [4] Jinxing Li[†], **Chuhao Zhou**[†], Xiaoqiang Ji, Mu Li, Guangming Lu, Yong Xu, and David Zhang. Multi-view Instance Attention Fusion Network for Classification. *Information Fusion*. († means 'Equal Contribution'.) 2023.

RESEARCH EXPERIENCE

MLLM for Robotic Perception and Planning | NTU | PhD Candidate

Aug 2024 – Present

Advisor: Jianfei Yang, assistant professor at School of Mechanical and Aerospace Engineering, NTU

- Propose HoloLLM, the first work to align MLLM with rare sensing modalities to achieve seamless human perception and reasoning with language grounded.
- Propose Universal Modality-Injection Projector (UMIP) and modality-specific encoders to deal with the data-scarce feature learning and modality-text alignment, respectively. (Paper accepted by *NeurIPS'25*.)
- Further improved the SAADG by enhancing the body parts features and designing a novel identification strategy.

Video-based Cross-modal Person Re-ID | HITSZ | Researcher

May 2022 – Jan 2024

Advisor: Jinxing Li, associate professor at School of Computer Science, HITSZ

Guangming Lu, professor and associate dean at School of Computer Science, HITSZ

- Created a large-scale video-based cross-modal person Re-ID dataset named HITSZ-PVCM, alleviated the overfitting problem in the Re-ID task and boosted performance through fine-grained details enhancement.
- Proposed a novel method named SAADG to simultaneously alleviate the intra- and inter-modal discrepancies by an adversarial strategy from the novel style disturbance perspective, and designed a graph-based interaction module to further capture the cross-view and cross-modal complementary information. I proposed the ideas of the paper and carried out most of the experiments. (Paper accepted by *MM'23, Oral*.)

- Further improved the SAADG by enhancing the body parts features and designing a novel identification strategy.

Multi-view Representations Learning | HITSZ | Researcher

May 2021 – Apr 2022

Advisor: Jinxing Li, associate professor at School of Computer Science, HITSZ

Guangming Lu, professor and associate dean at School of Computer Science, HITSZ

- Designed a novel network to promote the task of multi-view representation learning, exploited the additionally valuable information by instance-based learning, and proposed a algorithm to select noise-free template instances.

TECHNICAL SKILLS

Standard English Tests: TOEFL: Total 105 (Reading 29, Listening 26, Speaking 23, Writing 27).

Coding & Frameworks: Proficient in utilizing PyTorch to build various fundamental models (ResNet, FPN network, U-net, and Vision Transformer, etc.) and establish the customized model architectures. Proficient in implementing various widely used machine learning algorithms in Python, such as K-means, DBSCAN, SVM, etc.

Academic Skills: Proficient in the utilization of LaTeX, scientific figures drawing through Visio, and visualization technologies including t-SNE, GradCam, etc.